

Implementation of an ML model for image classification:-

1. Classifying terrain types, to plan the path for the movement of the rover. It prevents from entering into hazardous zones like craters.
2. Obstacle avoidance such as rocks, slopes or trenches.
3. Classifying rock types to create sampling.
4. Recognize features that might indicate water presence or biosignatures.
5. Works together with spectroscopic analysis for classification of samples found on Mars.
6. Classification of sky conditions to check out for dust storms, clouds, etc.
7. Classify images from cameras that monitor the rover's wheels, solar panels, robotic arms, etc. to detect wear or damage of any space systems.